ABHISHEK RAGHUWANSHI

AI/ML Engineer | LLM Applications | Computer Vision | MLOps

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LinkedIn | Portfolio | GitHub

PROFESSIONAL SUMMARY

AI/ML Engineer from IIT Bombay with hands-on experience building production-grade LLM applications, computer vision systems, and scalable ML pipelines. Expertise in deploying end-to-end AI solutions from data processing to containerization. Proven track record of delivering high-impact ML systems serving 10,000+ users.

EDUCATION

Indian Institute of Technology (IIT) Bombay **Jul 2022 - Jul 2024**

Master of Technology (M.Tech.) - Technology and Development | CPI: 8.09/10

PROFESSIONAL EXPERIENCE

Industrial Research And Consultancy Centre | IIT Bombay - AI Research Associate **Feb 2025 - Present**

AI Research Associate at IIT Bombay working on **AI/ML and Generative AI solutions for government and international organizations**, delivering end-to-end AI products including data pipelines, statistical analysis, model development, and deployment of LLM and computer vision systems on **cloud platforms (AWS and GCP)**.

Tech Stack: Python, LangChain LangGraph, OpenAI API (GPT-4), FAISS, Vector Databases, Prompt Engineering, LLaMA, HuggingFace, PyTorch, EfficientNet, OpenCV, Pandas, NumPy, Scikit-learn, Albumentations , Docker, Streamlit, Google Cloud Platform (GCP), AWS

World Food Programme (WFP), India

Key Contributions

- Led **end-to-end development and deployment of AI/ML digital solutions**, translating research requirements into scalable data science and Generative AI systems.
- Delivered **3+ AI/ML Proof-of-Concepts (PoCs)** and **1 production-grade document intelligence system** enabling automated research analysis and knowledge retrieval.
- Built **RAG-based document intelligence and semantic search pipelines** for large research datasets, enabling automated extraction and structured analysis of academic literature.
- Developed **LLM-driven data extraction and summarization workflows** supporting evidence synthesis and policy-level reporting.
- Reduced manual literature review effort by **~60%** and achieved **>85% retrieval accuracy** through optimized embeddings and retrieval pipelines.

Ministry of Agriculture and Farmers Welfare

Key Contributions

- Led **end-to-end development of AI-driven advisory and analytics systems**, including data preprocessing, model training, experimentation, and deployment.
- Delivered **5+ AI/ML Proof-of-Concepts (PoCs)** and **2 production-grade AI system** enabling automated knowledge retrieval from agricultural advisory documents and binary image classification system
- Built **RAG-based conversational AI and vector search pipelines** for semantic retrieval from large unstructured document repositories.
- Developed **deep learning computer vision models for agricultural image classification**, designing ML pipelines including EDA, dataset balancing, training, and evaluation.
- Implemented **MLOps workflows including containerization, CI/CD pipelines, and scalable cloud deployment** for AI applications.
- Achieved **~85% reduction in query response time**, **~92% retrieval relevance**, and **96% F1-score on 100K+ agricultural images** through model optimization and improved data pipelines.

iNDX.Ai | Full Stack Developer **Jul 2024 – Dec 2024**

Full Stack Developer building **scalable web applications and backend data systems**, designing APIs and database pipelines to support enterprise platforms.

Key Contributions

- Developed **full-stack applications and RESTful APIs** enabling scalable backend services and automated workflows.
- Designed **database schemas and backend data pipelines** for efficient data processing and management.
- Built **dynamic frontend interfaces** supporting authentication, role-based access control, and real-time interactions.
- Collaborated in **agile development environments using Git-based workflows and CI/CD practices**.

Tech Stack: Node.js, Express.js, PostgreSQL, Sequelize, Svelte, JavaScript, Git, Bitbucket, REST APIs

KEY PROJECTS

Twitter Sentiment Analysis with Deep Learning | IIT Bombay **Apr 2023 - Jun 2023**

- Developed sentiment classifier for brand monitoring processing **99,000+** tweets with **78%** accuracy using **LSTM + GloVe** embeddings, outperforming **Multinomial Naive Bayes baseline (76.3%)**
- Implemented robust preprocessing pipeline handling emojis, hashtags, URLs, and slang, **reducing noise by 35%** and improving model **convergence speed by 40%**
- Conducted **ablation study** comparing LSTM, BiLSTM, and CNN architectures with different embedding strategies (**Word2Vec, GloVe, FastText**)
- **Tech Stack:** Python, TensorFlow, LSTM, GloVe, NLTK, FastAPI, Pandas

Stock Price Movement Prediction using LSTM | IIT Bombay **Jan 2023 - May 2023**

- Implemented **time-series forecasting** models comparing LSTM, SVR, Ridge Regression, and Random Forest on **NIFTY-50** historical data using **Yahoo Finance API**
- Achieved best **R^2 of 0.84** through LSTM with **hyperparameter tuning (learning rate, dropout, layers)** and walk-forward validation to prevent data leakage
- Performed feature engineering creating technical indicators (**RSI, MACD, Bollinger Bands**) and evaluated models using **MAE, RMSE, NRMSE** metrics
- **Tech Stack:** Python, TensorFlow, Keras, Scikit-learn, yFinance, Pandas

LEADERSHIP & IMPACT

Associate Placement Manager (Initiatives) | IIT Bombay Placement Team **Sep 2022 - Dec 2022**

Led cross-functional team of 300+ members managing placements for 4,100+ students, demonstrating strong project management and stakeholder coordination:

- Achieved **14% higher Day-1 internship** selections by redesigning interview allocation model, saving **150+ man-hours**
- Deployed centralized resume verification portal **improving security** and user experience; **automated shortlist tracking**
- Designed **sector-specific** mock assessment program with **80%+ positive feedback** & **68% increase** in student participation
- Received **Certificate of Excellence** from Placement Office for outstanding contribution

ACHIEVEMENTS & INTERESTS

Certifications & Awards:

- Certificate of Excellence for Associate Placement Manager tenure - IIT Bombay Placement Office
- Gold Medal in Carrom - Hostel General Championship 2022, IIT Bombay

Areas of Interest: Large Language Models (LLMs), Generative AI, RAG Systems, Computer Vision, MLOps, Model Explainability, Agentic AI, Transformers, Production ML Systems

Soft Skills: Stakeholder Communication, Data Storytelling, Cross-functional Collaboration, Business Acumen

Hobbies: Reading AI/ML research papers, Playing cricket and carrom